# Advanced Artificial Intelligence: Diffusion Models

CSCI 546

## Instructor Info


Dr. Michael Wojnowicz (Mike)

Office Hrs: Tues 12:15-1:15, Thurs 12-2

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## Course Info

Prerequisites: Ability to code in some language (ideally Python). Integral and Differential Calculus. Previous experience with or strong interest in probability.

Class meetings: Tues, Thurs 10:50-12:05p

Wilson 1123

## Course description

This course provides an in-depth study of diffusion-based generative models in machine learning, including score-based models, denoising diffusion probabilistic models (DDPMs), and conditional and spatiotemporal variants. Students will explore the theoretical foundations, including Markov chains, stochastic differential equations, and score matching. Students will also learn how transformers can be used within diffusion modeling. Practical aspects of diffusion modeling will be explored in a final project.

## Readings

- *First half of the course*: Here we will focus on mathematics behind diffusion modeling. We will use the textbook:
  Grimmett, G., & Stirzaker, D. (2020). *Probability and random processes.* (4th Edition). Oxford university press.
- *Second half of the course:* Here we will focus on the machine learning of diffusion modeling, and will use the various resources given in the Class Schedule.

## Learning Outcomes

- To develop fluency in the mathematics underlying diffusion modeling (probability, Markov chains, and diffusion processes).
- To understand the fundamentals of diffusion modeling in machine learning;
- To apply diffusion modeling to a real or simulated dataset.

## Learning Philosophy

This course will use an "inquiry-based learning" approach. This means that classes will not be lecture-oriented. Instead, to quote Dr. Dana Ernst,

> *You will be expected to work actively to construct your own understanding of the topics at hand with the readily available help of me and your classmates. Many of the concepts you learn and problems you work on will be new to you and ask you to stretch your thinking. You will experience frustration and failure before you experience understanding. This is part of the normal learning process. If you are doing things well, you should be confused at different points in the semester.*

Active learning has been shown to increase student performance in STEM courses (e.g. see [1, 2]).

## Class activity

A typical class meeting will be structured (approximately) as follows:

Mathematics Module
- Review prev. exercises: 30 mins
- Mini-lecture: 15 mins
- New group exercises: 30 mins

ML Module
- Reading survey: 10 mins
- Peer lightning pres: 10 mins
- Good / bad / ugly / beautiful: 25 mins
- Mike's presentation: 25 mins

### Daily readings

Before each class meeting, you will be assigned a reading (or task). These tasks are listed on the class schedule at the end of the syllabus. The reading serves as preparation for actively participating in the class that day.

### Group exercises

For each class meeting within the mathematics module, students will be split randomly into groups of three to work on problems using the whiteboards. Mathematics is not a spectator sport!

## Grading

Course grades will be determined by the decomposition:

- Midterm: 30%
- Group Exercise Participation: 10%
- Reading surveys: 20%
- Presentations: 15%
- Project: 25%

Letter grades will be assigned as follows:
A: 93-100, A-: 90-93, B+: 87-90, B: 83-87, B-: 80-83, C+: 77-80, C: 73-77, C-: 70-73, D+: 67-70, D: 63-67, D-: 60-63, F: 0-60.

Many products (midterm questions, reading surveys, lightning chats) will be graded as follows:

- 4: Clear, correct response demonstrating strong understanding of the concept behind the question.
- 3: Mostly correct with reasonable explanation, but with minor mathematical or conceptual issues.
- 2: Shows partial understanding or some useful intuition, but contains significant errors or gaps in reasoning.
- 1: Minimal understanding demonstrated; response is incomplete, unclear, or only shows limited engagement with the question.
- 0: No meaningful work shown or question left blank.

**Reading surveys.** The purpose of reading surveys is to check that you did the reading and tried to understand it. Questions will tend towards being open-ended, and will be graded as above.

**Makeup policy.** It is highly recommended to keep up with the class, as the material is cumulative. However, I drop 2 participation days with no consequence. Missed material beyond that (with a valid explanation and documentation) can be replaced with by increasing the weight of the midterm and final project grades.

**Diversity and Inclusivity Statement.** I consider this classroom to be a place where you will be treated with respect, and I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, ability - and other visible and non-visible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. Your suggestions about how to improve the value of diversity in this course are encouraged and appreciated. Please let me know ways to improve the effectiveness of the course for you personally or for other students or student groups.

**Accommodations for Students with Disabilities.** If you are a student with a disability and wish to use your approved accommodations for this course, contact me during my office hours to discuss. Please have your Accommodation Notification available for verification of accommodations. Accommodations are approved through the Office of Disability Services located in 137 Romney Hall. www.montana.edu/disabilityservices.

**Student Conduct.** You are expected to abide by MSU's Code of Student Conduct.

**Scholarly Responsibilities.** The class has no homework. However, to be successful in the course, you need to do the following:

- Before class: *Read the assigned section.* Be sure to read actively , with a pencil and paper in hand. For the mathematics module, work out some of the examples within the section on your own, using the textbook to check your work.
- During class: *Attend class.* This allows you to participate in the group activities and discussions.
- After class: *Finish working through the group exercises.* Solutions to group exercises are posted in Canvas. During class time, you will probably just get the gist of some problems. After class, you should make sure you know how to do all of the group exercises.

Class Schedule

# MODULE 0: Course Overview

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|---|---|---|---|
| Tues | Jan 13 | Course Overview | |

# MODULE 1: Mathematics

## MODULE 1A: Probability Fundamentals

| | | | |
|---|---|---|---|
| Thurs | Jan 15 | Events and their probabilities | GS Sec 1.1-1.7 |
| Tues | Jan 20 | Random variables and their distributions | GS Sec 2.1-2.3 |
| Thurs | Jan 22 | Random variables and their distributions | GS Sec 2.4-2.6 |
| Tues | Jan 27 | Discrete random variables | GS Sec 3.1-3.5 |
| Thurs | Jan 29 | Discrete random variables | GS Sec 3.6-3.10 |
| Tues | Feb 3 | Continuous random variables | GS Sec 4.1-4.5 |
| Thurs | Feb 5 | Continuous random variables | GS Sec 4.6-4.11 |

## MODULE 1B: Markov Chains

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|---|---|---|---|
| Tues | Feb 10 | Markov Chains | GS Sec 6.1, 6.2 |
| Thurs | Feb 12 | Markov Chains | GS Sec 6.3, 6.4 |
| Tues | Feb 17 | Markov Chains | GS Sec 6.5, 6.6, 6.9 |

## MODULE 1C: Diffusion Processes

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|---|---|---|---|
| Thurs | Feb 19 | Brownian motion | Chang Sec 5.1-5.6. Suppl: GS Sec 8.1, 8.5 |
| Tues | Feb 24 | Diffusion Processes | Chang Sec 6.1-6.5. Suppl: GS Sec 13.3 |
| Thurs | Feb 26 | Stochastic Calculus | Chang Sec 6.7-6.8. Suppl: GS Sec 13.7 |
| Tues | Mar 3 | Itô integral | Chang Sec 6.9-6.11. Suppl: GS Sec 13.8-13.9 |

## MIDTERM

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|---|---|---|---|
| Thurs | Mar 5 | MIDTERM EXAM | |

# MODULE 2: Machine Learning

## MODULE 2A: Introduction to Diffusion

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|---|---|---|---|
| Tues | Mar 10 | Variational Inference | Blei, D. M., Kucukelbir, A., & McAuliffe, J. D. (2017). Variational inference: A review for statisticians. *Journal of the American statistical Association*, 112(518), 859-877. |
| Thurs | Mar 12 | Variational Autoencoders | Kingma, D. P., & Welling, M. (2014). Auto-encoding variational Bayes. In *Proceedings of the International Conference on Learning Representations (ICLR)*. |
| | | No Classes — Spring Break | |
| Tues | Mar 24 | Classic Diffusion Model | Sohl-Dickstein, J., Weiss, E., Maheswaranathan, N., & Ganguli, S. (2015, June). Deep unsupervised learning using nonequilibrium thermodynamics. In *International conference on machine learning* (pp. 2256-2265).<br><br>*Lightning chat*: Josh B. & Racquel |
| Thurs | Mar 26 | Overview of Diffusion Models | Christopher Bishop (2023), *Deep Learning* (Chapter 20: Diffusion Models). Note: for now, only read Secs 20.1 & 20.2.<br><br>*Lightning chat*: Blake & Matt |
| Tues | Mar 31 | Diffusion exercise | Code up a simple diffusion model!<br><br>*Reference:* https://e-dorigatti.github.io/math/deep%20learning/2023/06/25/diffusion.html |

## MODULE 2B: Diffusion and Score Matching

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|---|---|---|---|
| Thurs | Apr 2 | Classic Score Matching | Hyvärinen, A. (2005). Estimation of non-normalized statistical models by score matching. *Journal of Machine Learning Research*, 6(4).<br><br>Supplemental Exercises: https://ericmjl.github.io/score-models/notebooks/01-score-function.html<br><br>Supplemental Reading: Christopher Bishop (2023), *Deep Learning* Sec 20.3<br><br>*Lightning chat*: Sam & Dillon |
| Tues | Apr 7 | Diffusion with Score Matching | Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33, 6840-6851.<br><br>Supplemental topic: Diffusion and variational inference. Kingma, D., & Gao, R. (2023). Understanding diffusion objectives as the elbo with simple data augmentation. *Advances in Neural Information Processing Systems*, 36, 65484-65516<br><br>*Lightning chat*: Jacob & Mike |

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| Thurs | Apr 9 | Diffusion and SDEs | Song, Y., Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., & Poole, B. (2021). Score-based generative modeling through stochastic differential equations. In *Proceedings of the International Conference on Learning Representations (ICLR).*<br><br>*Lightning chat*: Owen, Ismoiljon |

## MODULE 2C: Diffusion with Transformers

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|---|---|---|---|
| Tues | Apr 14 | Intro to Transformers | Christopher Bishop (2023), *Deep Learning* (Chapter 12: Transformers).<br><br>Supplemental: Sasha Rush's *LLMs in 5 formulas*<br><br>*Lightning chat*: Austin, Logan, Tatiana |
| Thurs | Apr 16 | Attention Exercises | https://classic.d2l.ai/chapter_attention-mechanisms/index.html |
| Tues | Apr 21 | Diffusion with transformers | Peebles, W., & Xie, S. (2023). Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 4195-4205).<br><br>*Lightning chat: Diego, Aubrey* |

## MODULE 2D: Spatiotemporal Diffusion

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|---|---|---|---|
| Thurs | Apr 23 | Conditional score-based diffusion | Tashiro, Y., Song, J., Song, Y., & Ermon, S. (2021). Csdi: Conditional score-based diffusion models for probabilistic time series imputation. *Advances in neural information processing systems*, 34, 24804-24816.<br><br>*Lightning chat: Josh S, Josh C.* |
| Tues | Apr 28 | Probabilistic weather forecasting | Price, I., Sanchez-Gonzalez, A., Alet, F., Andersson, T. R., El-Kadi, A., Masters, D.... & Willson, M. (2025). Probabilistic weather forecasting with machine learning. *Nature*, 637(8044), 84-90.<br><br>*Lightning chat: Lina, Laura* |

## MODULE 2E: Projects

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| Thur | Apr 30 | Final project workshop |
| Thurs | May 7 | Project Submission Deadline |

Note: The class schedule is tentative; it is subject to change as the course progresses.

## * References


[1] Louis Deslauriers, Ellen Schelew, and Carl Wieman. Improved learning in a large-enrollment physics class. *science*, 332(6031):862–864, 2011.

[2] Scott Freeman, Sarah L Eddy, Miles McDonough, Michelle K Smith, Nnadozie Okoroafor, Hannah Jordt, and Mary Pat Wenderoth. Active learning increases student performance in science, engineering, and mathematics. *Proceedings of the national academy of sciences*, 111(23):8410–8415, 2014.